

Blockchain: Smart Contracts

Lecture 9

Stablecoins & Lending Protocols

Recap: Solidity

Everything is a contract:

- Contracts manage state variables
- Contracts have functions that can be called externally
- Can inherit code from other contracts (`contract A is B,C`)

Global objects: `block`, `tx`, `msg`

(e.g. `block.number`, `tx.gasprice`, `msg.sender`)

An example: ERC20 tokens

- <https://github.com/ethereum/EIPs/blob/master/EIPS/eip-20.md>
- A standard API for fungible tokens. (ERC-721 for non-fungible tokens)
- An ERC20 token is itself a smart contract that maintains all user balances:
mapping(address => uint256) internal **_balances**;
- A standard interface allows other contracts to interact with every ERC20 token. No need for special logic for each token.

ERC20 token interface

function **transfer**(address _to, uint256 _value) external returns (bool);

function **transferFrom**(address _from, address _to, uint256 _value) external returns (bool);

function **approve**(address _spender, uint256 _value) external returns (bool);

function **totalSupply**() external view returns (uint256);

function **balanceOf**(address _owner) external view returns (uint256);

function **allowance**(address _owner, address _spender) external view returns (uint256);

An example ...

Consider two ERC-20 tokens: say USDC and WETH

- USDC is a contract that maintains a **_balances[]** mapping
- WETH is a different contract that also maintains **_balances[]**

Say Bob owns 5 USDC and 2 WETH. This is recorded as:

In USDC contract: **_balances[Bob's address] == 5**

In WETH contract: **_balances[Bob's address] == 2**

Wallet software shows all the coins associated with Bob's address

Anyone can read ERC20 _balances[]

Transaction Hash: 0x6b85ca95e484d94503d1276456bfc32cc55f6fdb8bb231ff83....

Tells the USDC contract to transfer 10,010.00 USDC
from Circle's account to 0x7656159E42209A95b77aD374d...

Storage Address: 0x4d3e7741e6c98c0c469419fcfe58fa7ec622d7b26345802d22d17415768760f8

Before:

Hex ▾

→ 0x00

After:

Hex ▾

→ 0x002540be400

recipient's
entry

Storage Address: 0x57d18af793d7300c4ba46d192ec7aa095070dde6c52c687c6d0d92fb8532b305

Before:

Hex ▾

→ 0x000266988cda8061

After:

Hex ▾

→ 0x0002669638ce9c61

Circle's
entry

(Circle's balance after)

Calling other contracts

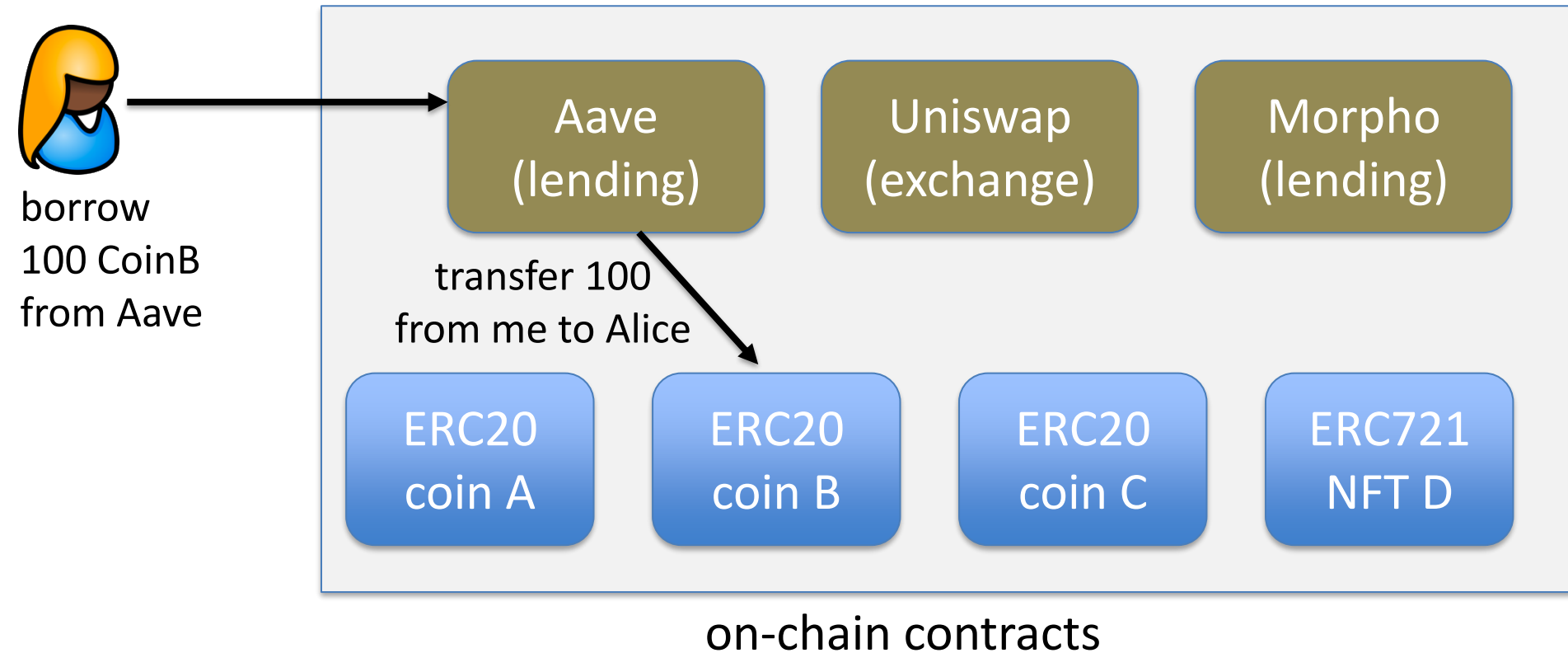
Addresses can be cast to contract types.

```
address _usdc = 0x7656159E42209A95b77aD374d...;  
ERC20Token usdcContract = ERC20Token(_usdc);
```

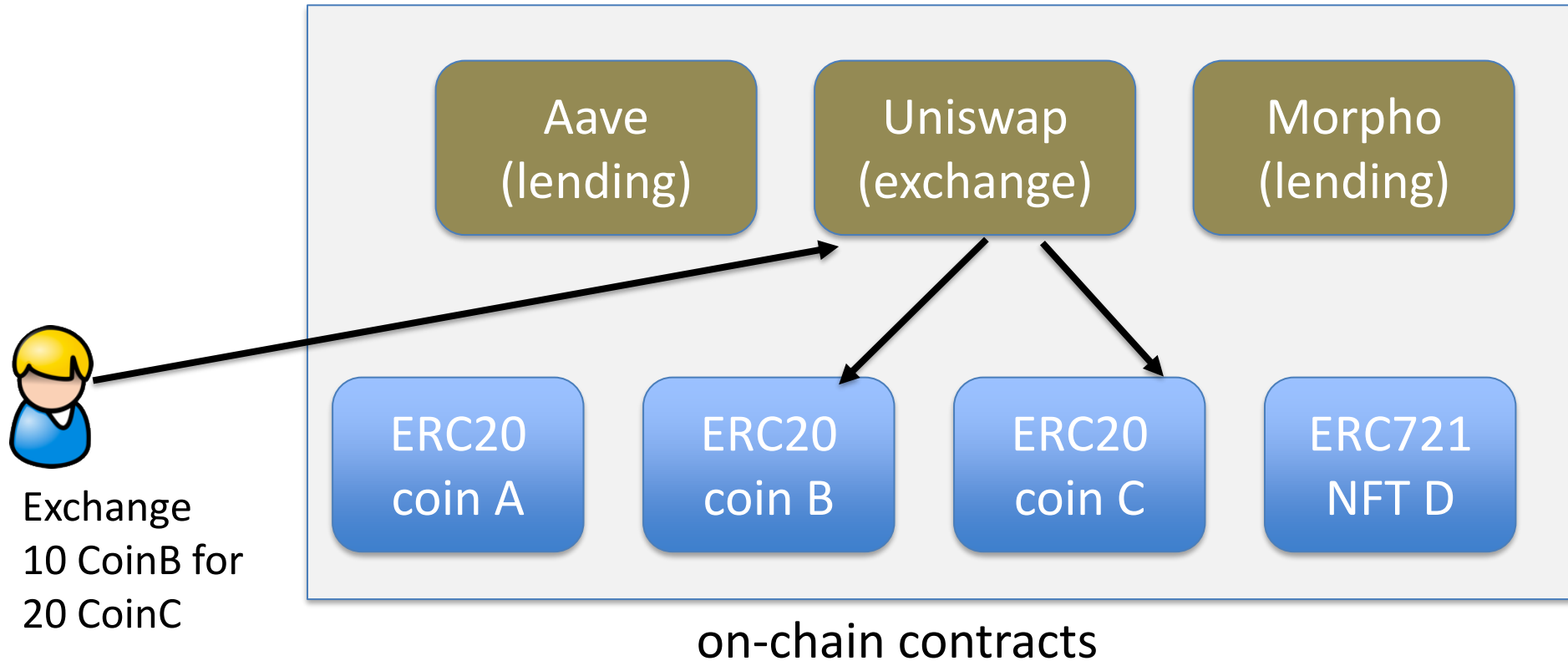
To call the “transfer” function of contract at address _usdc:

```
usdcContract.transfer(_to, _value);
```

The world of DeFi



The world of DeFi



DeFi app #1: Stablecoins

Stable Coins

A cryptocurrency designed to trade at a fixed price

- Examples: **1 coin = 1 USD**, 1 coin = 1 EUR, 1 coin = 1 USDX

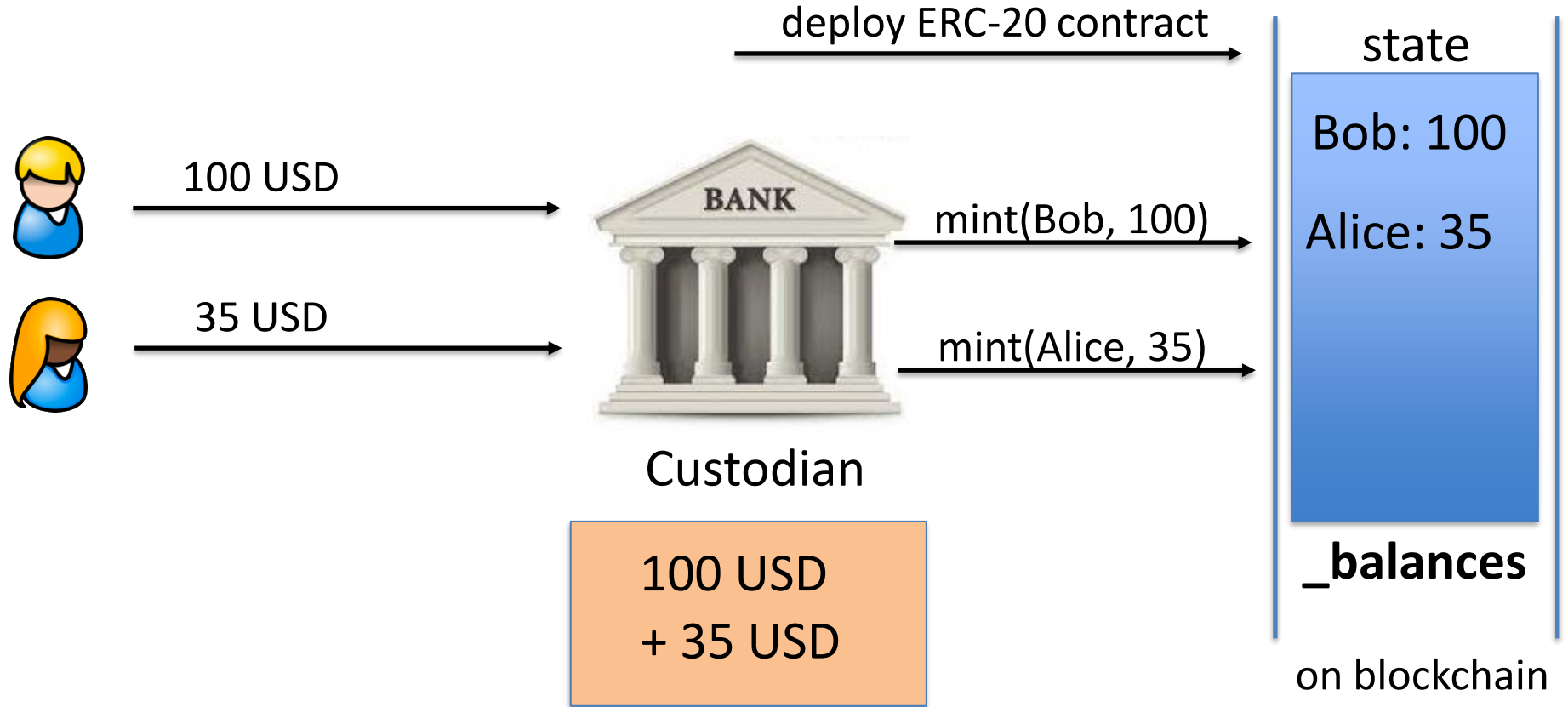
Goals:

- Integrate real-world currencies into on-chain applications
- Enable people without easy access to USD, to hold and trade a USD-equivalent asset

Types of stable coins

	centralized	algorithmic
collateralized	custodial stablecoins (USDC)	synthetics (DAI, RAI)
Un(der)collateralized	central bank (digital) currency	Undercollateralized stablecoins

Custodial stablecoins: minting



Custodial stablecoins: transfers



pay Carol 15\$:

transfer(Bob → Carol, 15)

(and gas fee)

Transfers are done on-chain
(custodian is not involved)

135 USD

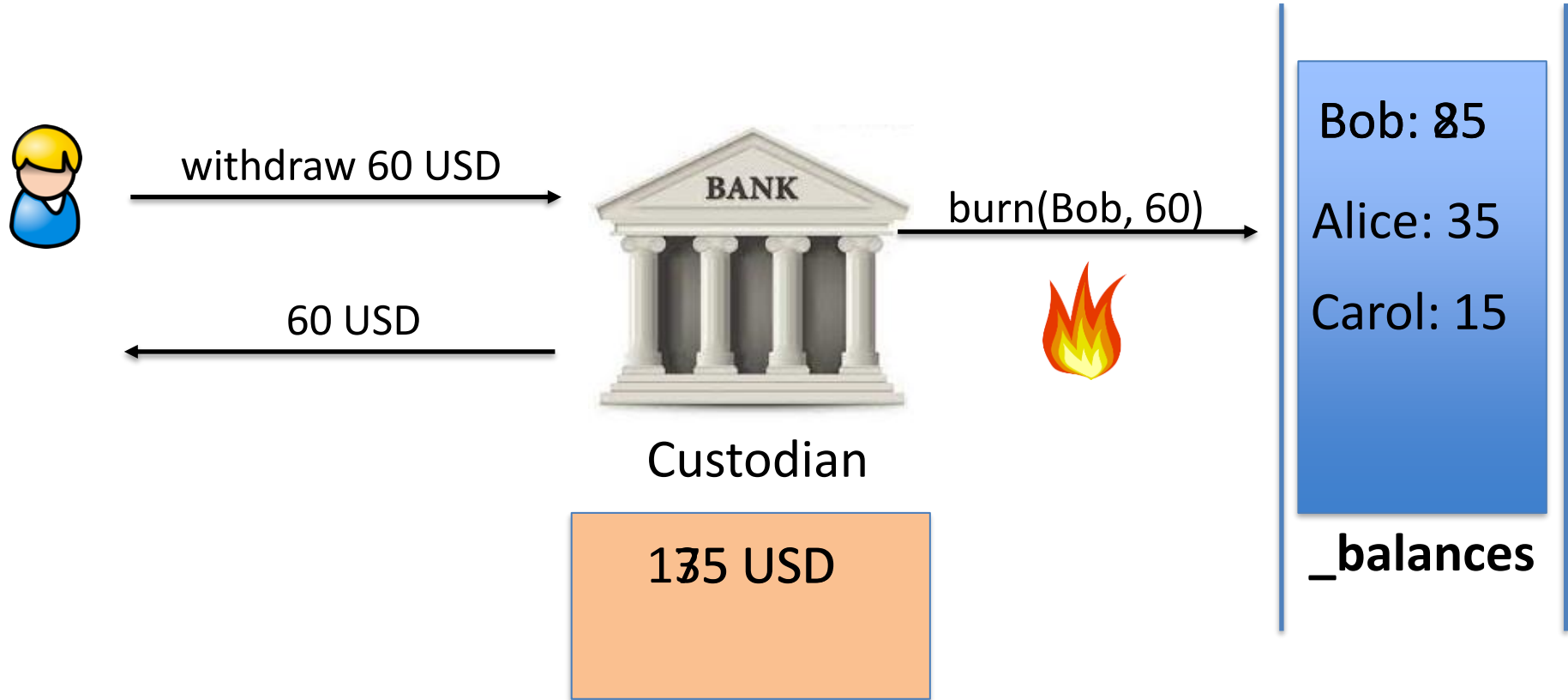
Bob: 800

Alice: 35

Carol: 15

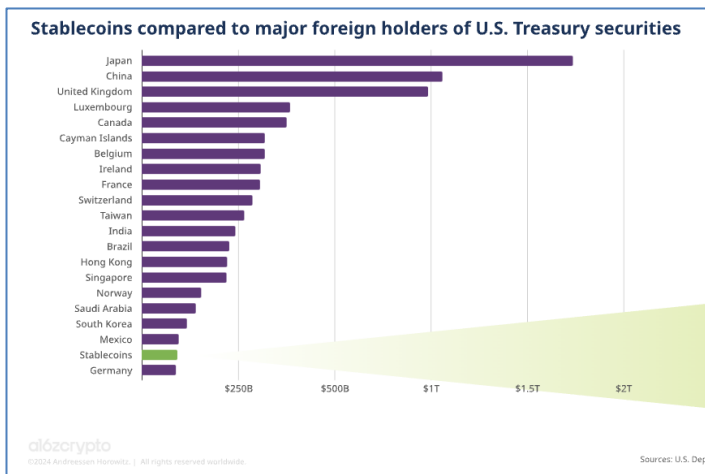
_balances

Custodial stablecoins: withdrawal



Two Examples

	Coins issued	24h volume
USDC	76 B	20.4 B
USDT	184 B	164.6 B



stablecoin providers
are a large holder
of U.S. debt.

Some issues

Custodian keeps treasury in a traditional bank

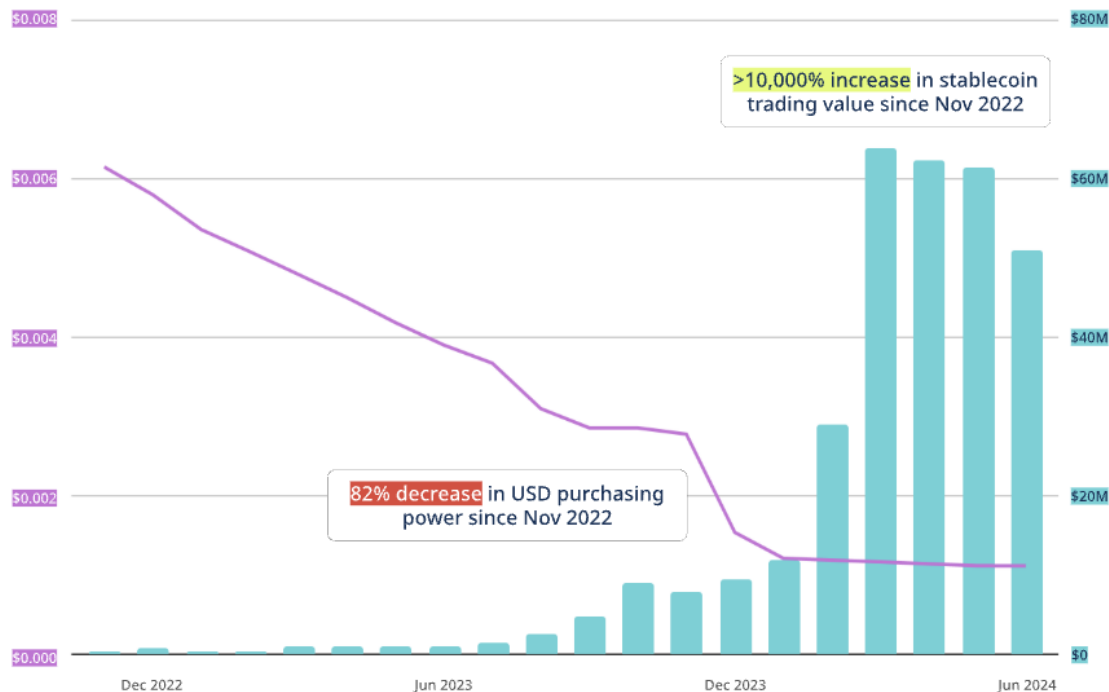
- Must be audited to ensure treasury is available
- Earns interest on deposits

Custodian has strong powers:

- Can freeze accounts / refuse withdrawal requests
- Custodian can remove funds from user balances

In countries with rampant inflation, people turn to stablecoins to protect their assets

Argentine peso (ARS) purchasing power in USD vs. stablecoin trading value with ARS on Bitso



Retail-sized stablecoin value (i.e. transactions under \$10,000) received in Argentina is growing at a faster rate than value received in any other crypto asset, including BTC, ETH, and altcoins. Bitso is a leading regional exchange in Latin America.

Source: Chainalysis Geography of Cryptocurrency Report (Oct 2024). Data is through 6/30/2024.

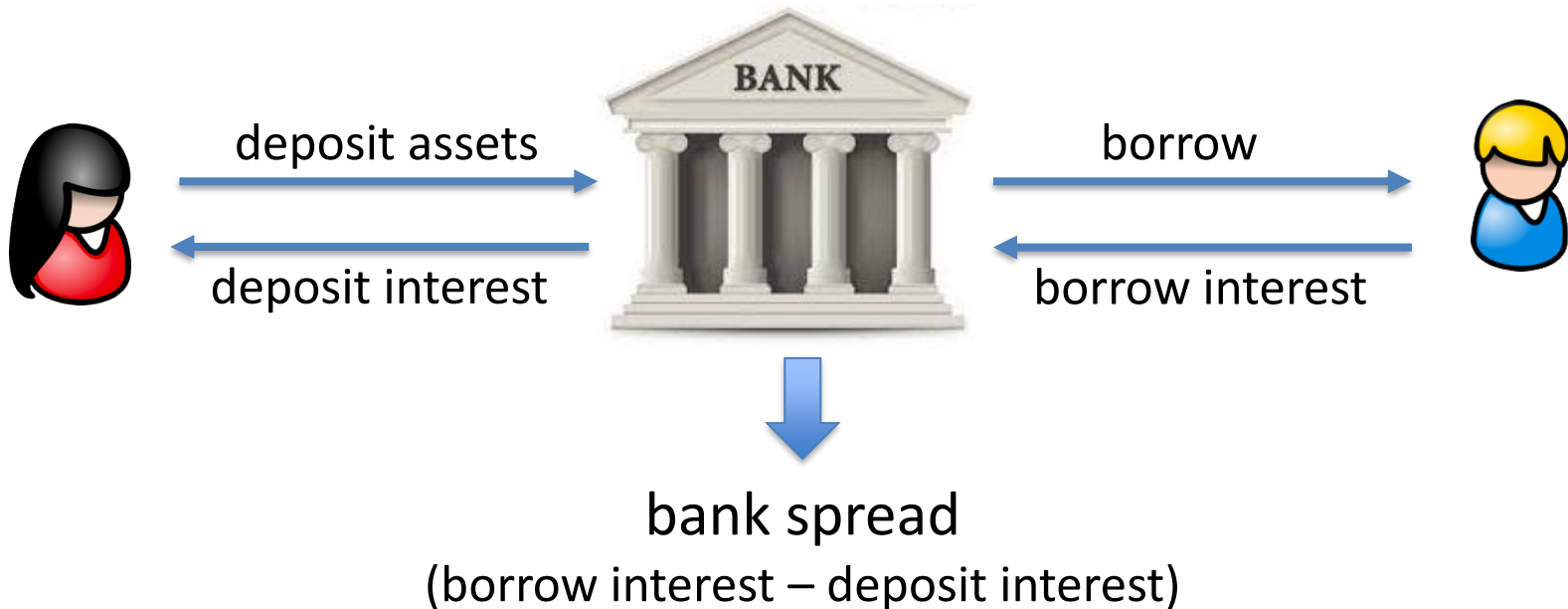
DeFi app #2: Lending Protocols

Goal: explain how decentralized lending works

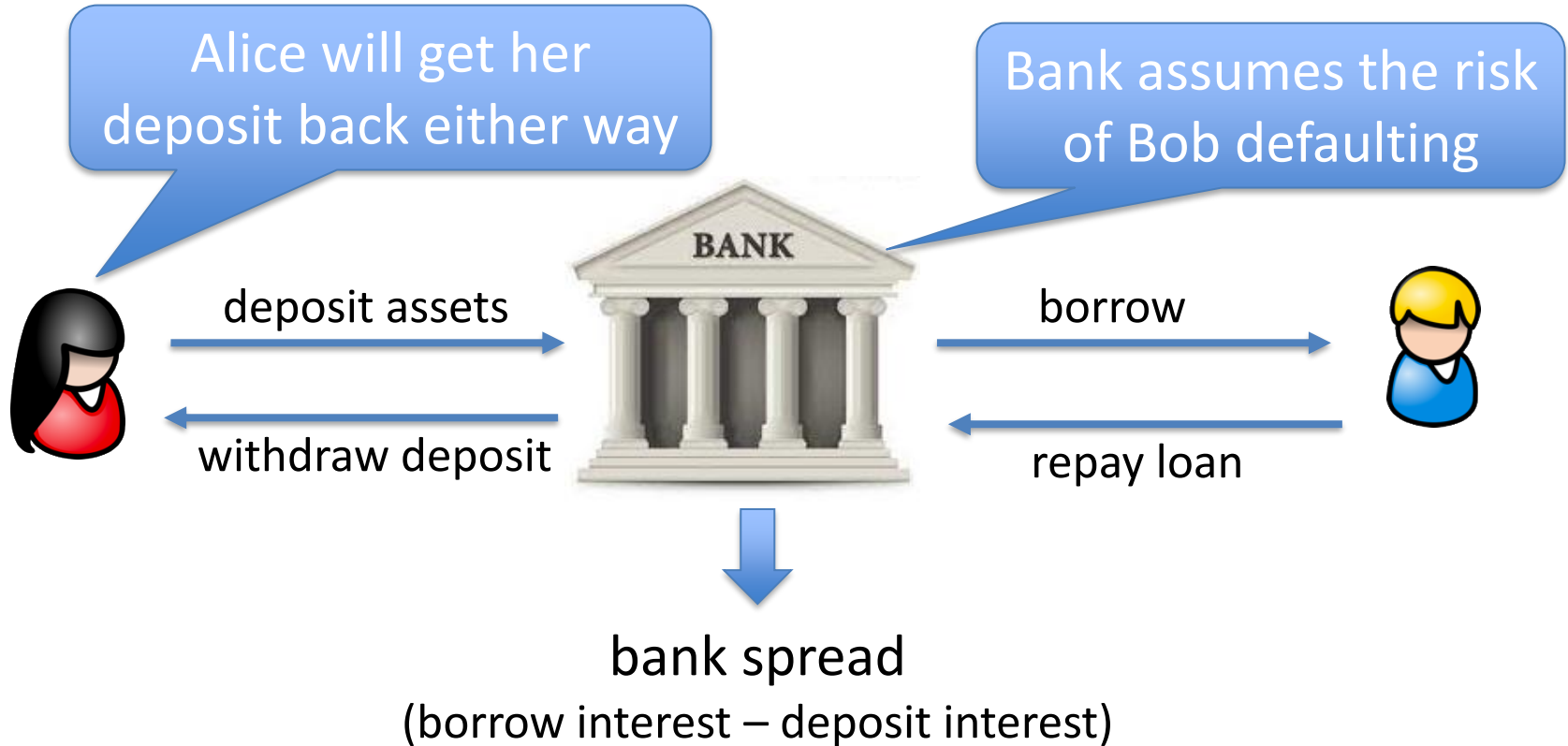
This is not investment or financial advice

The role of banks in the economy

Banks bring together lenders and borrowers

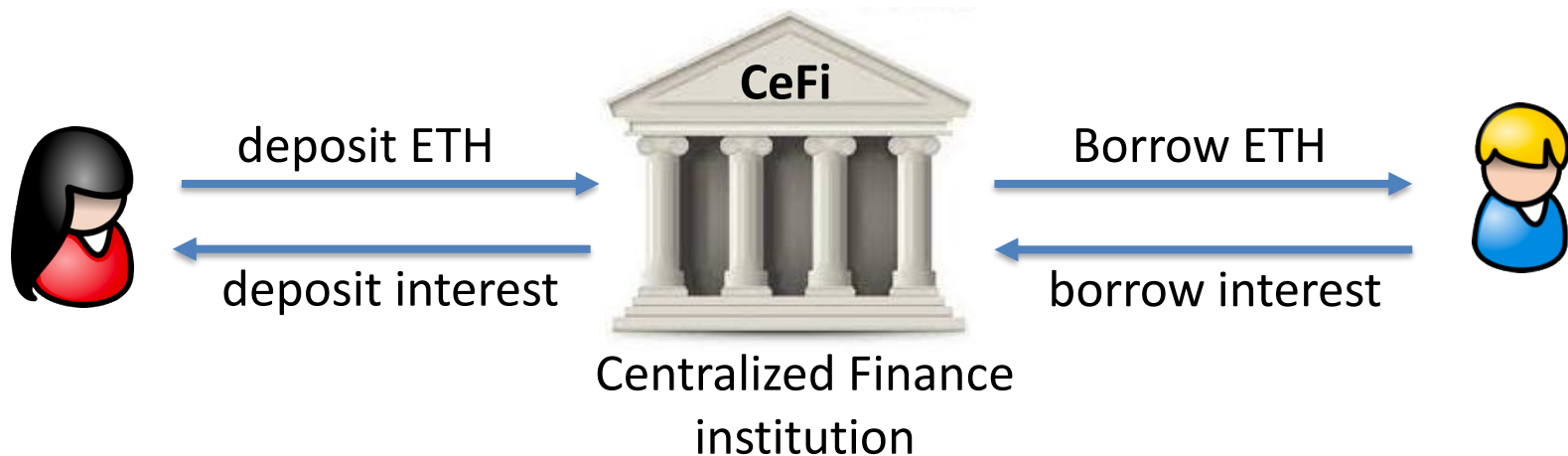


The role of banks in the economy



Crypto: CeFi lending (e.g., Nexo)

Same as with a traditional bank:



Alice gives her assets to the CeFi institution to lend out to Bob

The role of collateral

(1 ETH = 100 UNI)

CeFi's concern: what if Bob defaults on loan?

⇒ CeFi will absorb the loss

Solution: require Bob to lock up collateral

collateral



deposit 500 UNI

Borrow 1 ETH



debt position:

+ 500 UNI
- 1 ETH

interest deducted from collateral

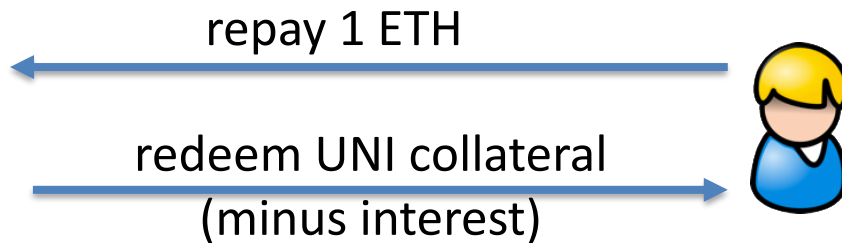
over collateralized loan

The role of collateral

Several things can happen next:

(1 ETH = 100 UNI)

(1) Bob repays loan



debt position:

~~+ 50 UNI~~
~~- 1 ETH~~

The role of collateral

Several things can happen next:

(1) Bob repays loan

(2) Bob defaults on loan

(1 ETH = 100 UNI)

Ok, I'll keep
(100 + penalty) UNI



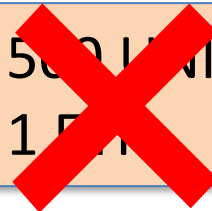
I can't repay 1 ETH

redeem remaining UNI collateral
(400 - interest - penalty) UNI



debt position:

+ 500 UNI
- 1 ETH



The role of collateral

Several things can happen next:

(1 ETH = 400 UNI)

(1) Bob repays loan

(2) Bob defaults on loan

(3) Liquidation: value of loan increases relative to collateral



I need to liquidate
your collateral
(and charge a penalty = 20 UNI)



debt position:

+ 80 UNI
- 0 ETH

lender needs to liquidate **before** $\text{value}(\text{debt}) > \text{value}(\text{collateral})$

Terminology

Collateral: assets that serve as a security deposit

Over-collateralization: to borrow, borrower has to provide

$$0.8 \times \text{value(collateral)} > \text{value(debt)}$$

Loan-to-Value (LTV) ratio

Liquidation Threshold

Liquidation:

if $0.83 \times \text{value(collateral)} < \text{value(debt)}$

then collateral is liquidated to pay debt until inequality flips
(liquidation reduces both sides of the inequality)

Loan-to-Value (LTV)

Loan-to-Value (LTV) $\in [0,1]$

- Max value that can be borrowed using this collateral
- High volatility asset $\Rightarrow$ low LTV
- Relatively stable asset $\Rightarrow$ higher LTV

Examples: (on Aave)

LTV: ETH: 80.5%, USDC: 75%, DAI: 63%

Liquidation th.: ETH: 83%, USDC: 78%, DAI: 77%

Health of a debt position

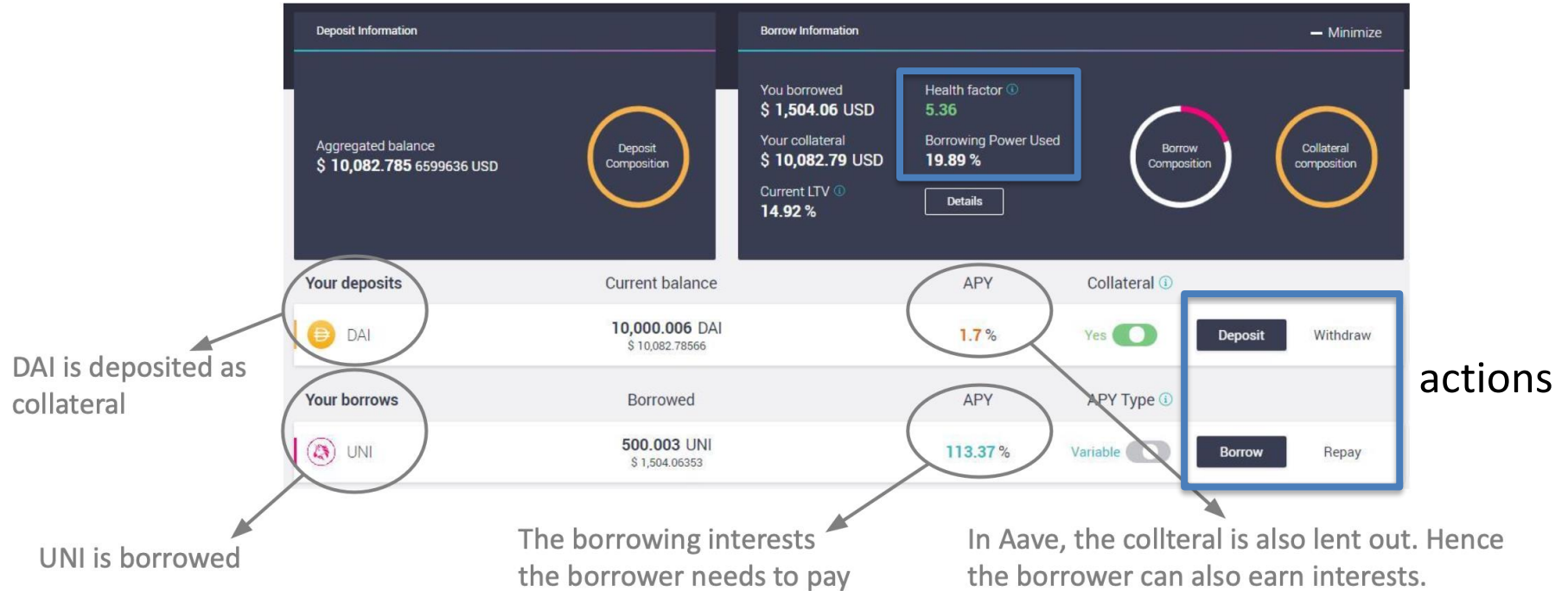
$$\text{BorrowCapacity} = \sum_i \text{value}(\text{collateral}_i) \times \text{LTV}_i$$

(in ETH)

$$\text{health} = \frac{\sum_i \text{value}(\text{Collateral}_i) \times (\text{Liquidation Threshold}_i)}{\text{value}(\text{TotalDebt})}$$

health < 1 $\Rightarrow$ triggers liquidation until (health $\geq$ 1)

Example: Aave dashboard (a DeFi lending Dapp)



Why borrow ETH?

If Bob has collateral, why can't he just buy ETH?

- Bob may need ETH (e.g., to buy in-game assets), but he might not want to sell his collateral (e.g., an NFT)
- As an investment strategy: using UNI to borrow ETH gives Bob exposure to both

The problem with CeFi lending

Users must trust the CeFi institution:

- Not to get hacked, steal assets, or miscalculate
- This is why traditional finance is regulated
- Interest payments go to the exchange, not liquidity provider Alice
- CeFi fully controls spread (borrow interest – deposit interest)

DeFi Lending

Can we build an on-chain lending Dapp?

⇒ no central trusted parties

⇒ code available on Ethereum for inspection

A first idea: an order book Dapp

Order Book Protocol

LENDERS



(large institutions, banks)



BORROWERS



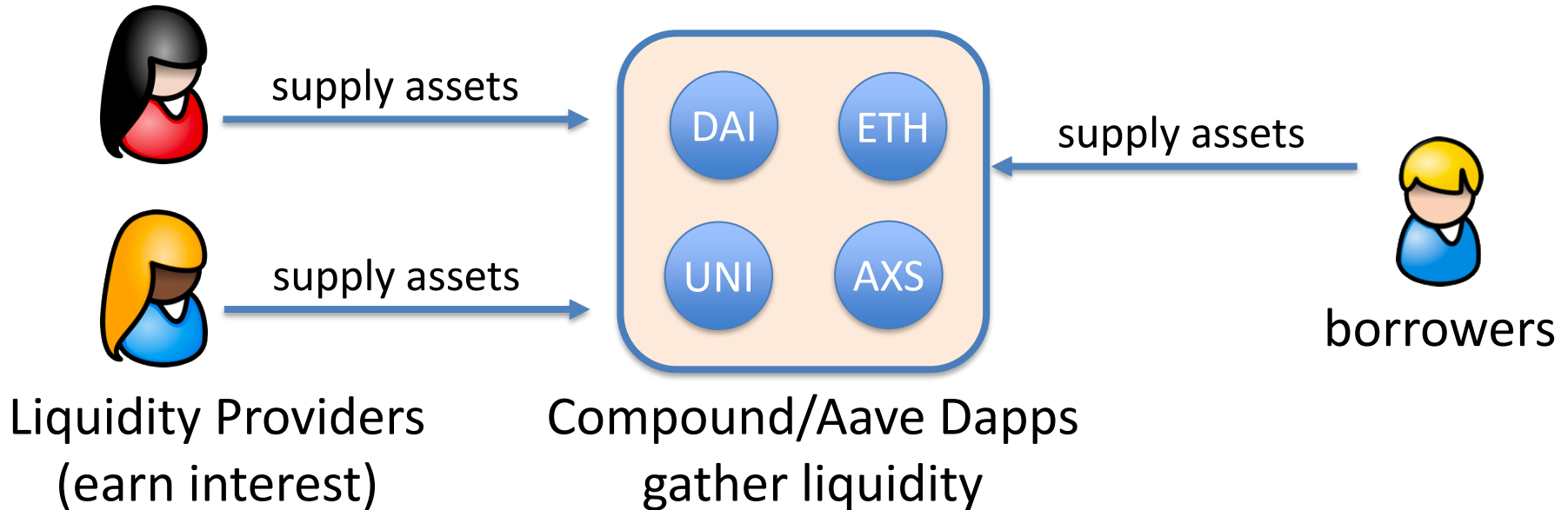
Credit: Eddy Lazzarin

Challenges

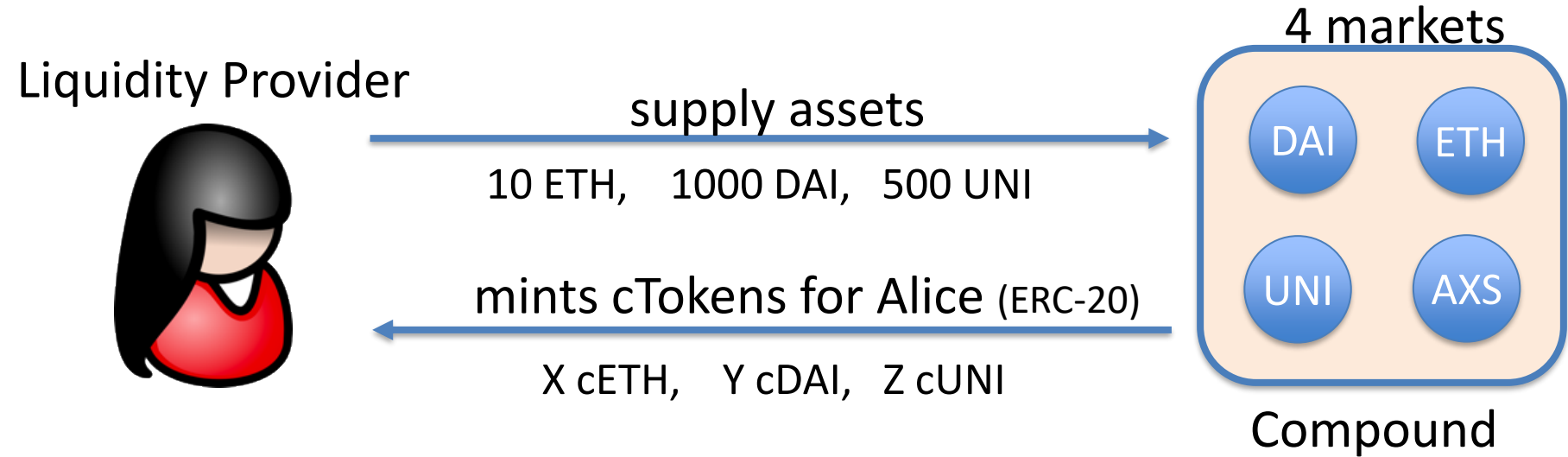
- **Computationally expensive:** matching borrowers to lenders requires many transactions per person (post a bid, retract if the market changes, repeat)
- **Concentrated risk:** lenders are exposed to their direct counterparty defaulting
- **Complex withdrawal:** a lender must wait for their counter-parties to repay their debts

A better approach: liquidity pools

Over-collateralized lending: Compound and Aave

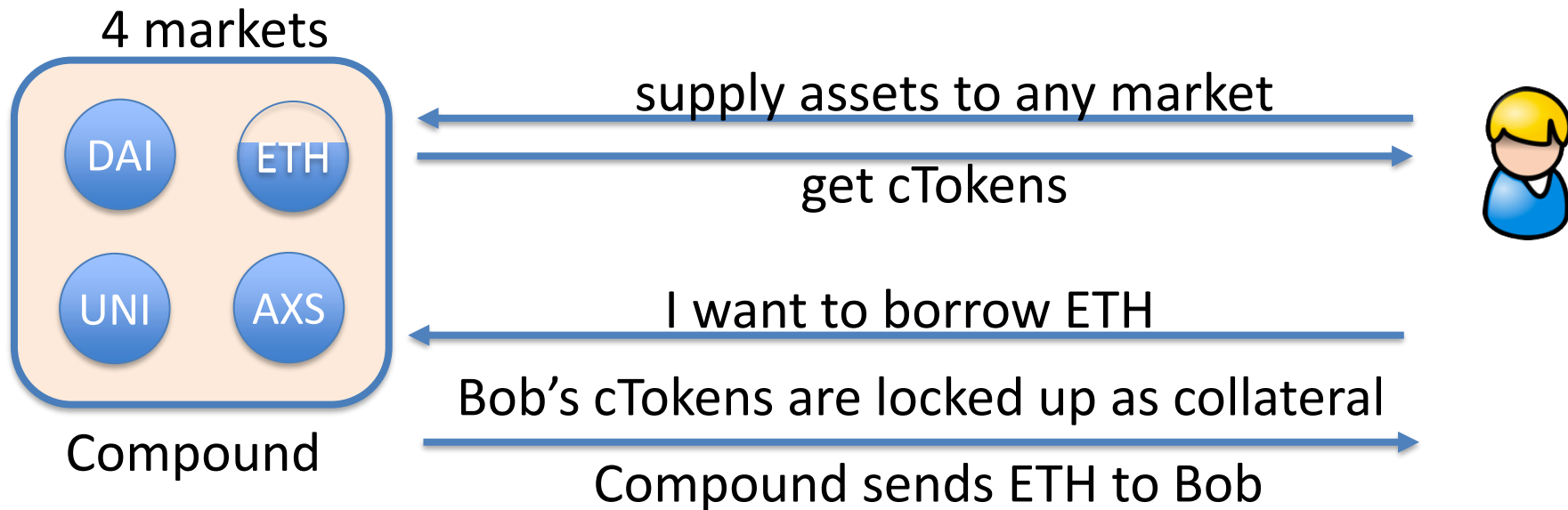


Example: Compound cTokens



Value of X, Y, Z is determined by the current exchange rate:
Token to cToken exchange rate is calculated every block

Borrowers



Bob's accrued interest increases ETH/cETH exchange rate

⇒ benefit cETH token holders (ETH liquidity providers)

The interest rate: constantly updates

Key idea: determined by demand for asset vs. asset market size

Utilization ratio: $U_{ETH} = \frac{\text{totalBorrowBalance}_{ETH}}{\text{availableBalance}_{ETH} + \text{totalBorrowBalance}_{ETH}}$

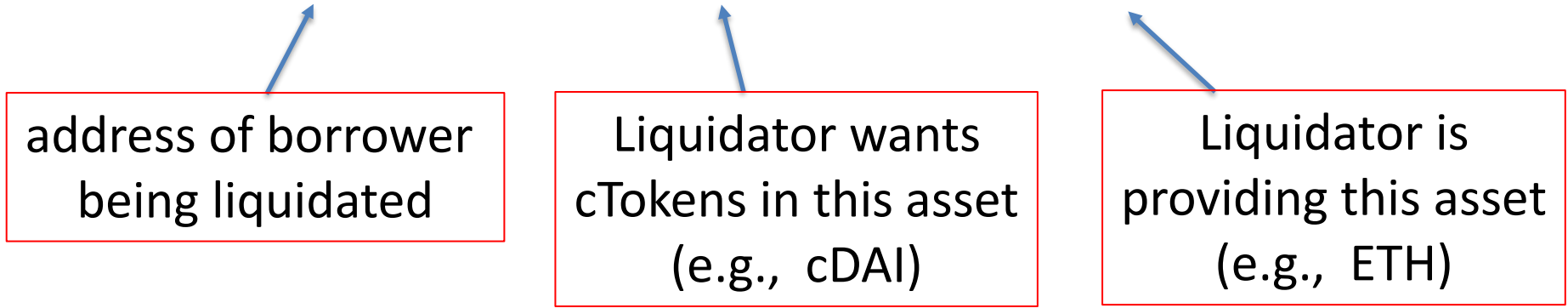
higher totalBorrowBalance, or
lower availableBalance in contract  higher $U_{ETH} \in [0,1]$

$$\text{interestRate}_{ETH} = \text{BaseRate}_{ETH} + U_{ETH} \times \text{slope}_{ETH}$$

Liquidation: $\text{debt} > \text{BorrowCapacity}$

If user's health < 1 then anyone can call:

liquidate(borrower, CollateralAsset, BorrowAsset, uint amount)



address of borrower
being liquidated

Liquidator wants
cTokens in this asset
(e.g., cDAI)

Liquidator is
providing this asset
(e.g., ETH)

This function transfers liquidator's ETH into ETH market,
and gives the liquidator cDAI from user's collateral

Liquidation: $\text{debt} > \text{BorrowCapacity}$

If user's health < 1 then anyone can call:

Liquidator is repaying the user's ETH debt
and getting the user's cDAI

[at a discounted exchange rate -- penalty for user]

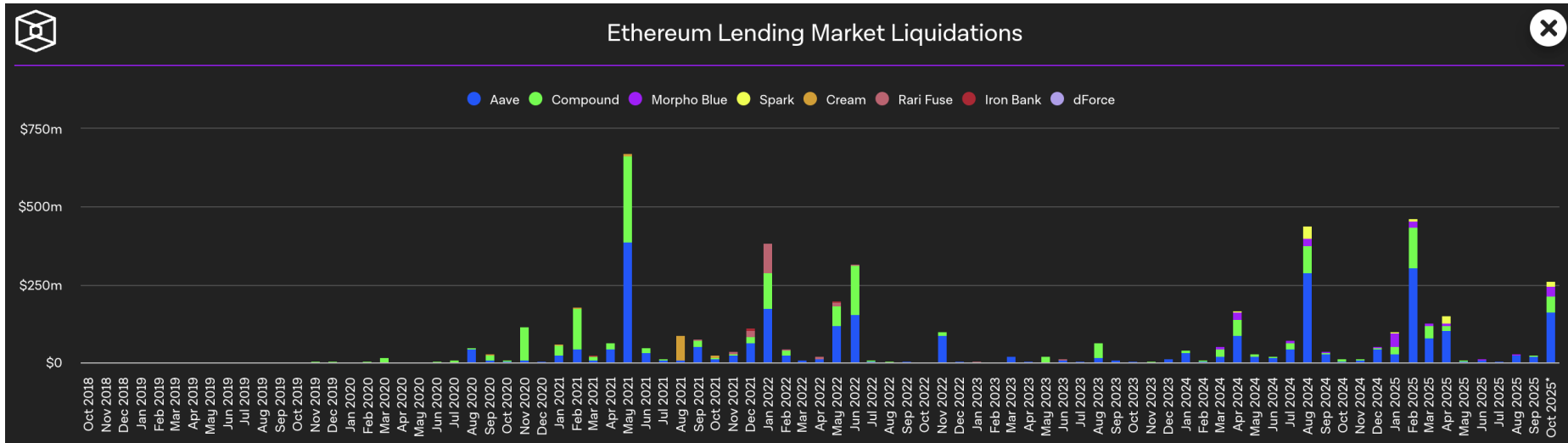
(e.g., cDAI)

(e.g., ETH)

This function transfers liquidator's ETH into ETH market,
and gives the liquidator cDAI from user's collateral

This happens ...

Liquidations on Ethereum:



Caused by collateral price drops or debt APR spikes

What is liquidation risk?

Historical UNI interest rate on Aave (APR):

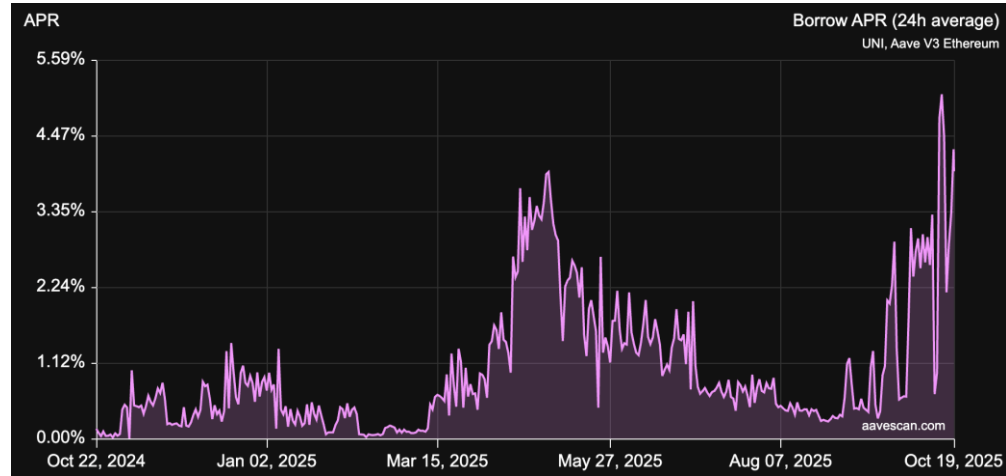
Demand for UNI spikes

⇒ price of UNI spikes

⇒ user's debt shoots up

⇒ user's health drops

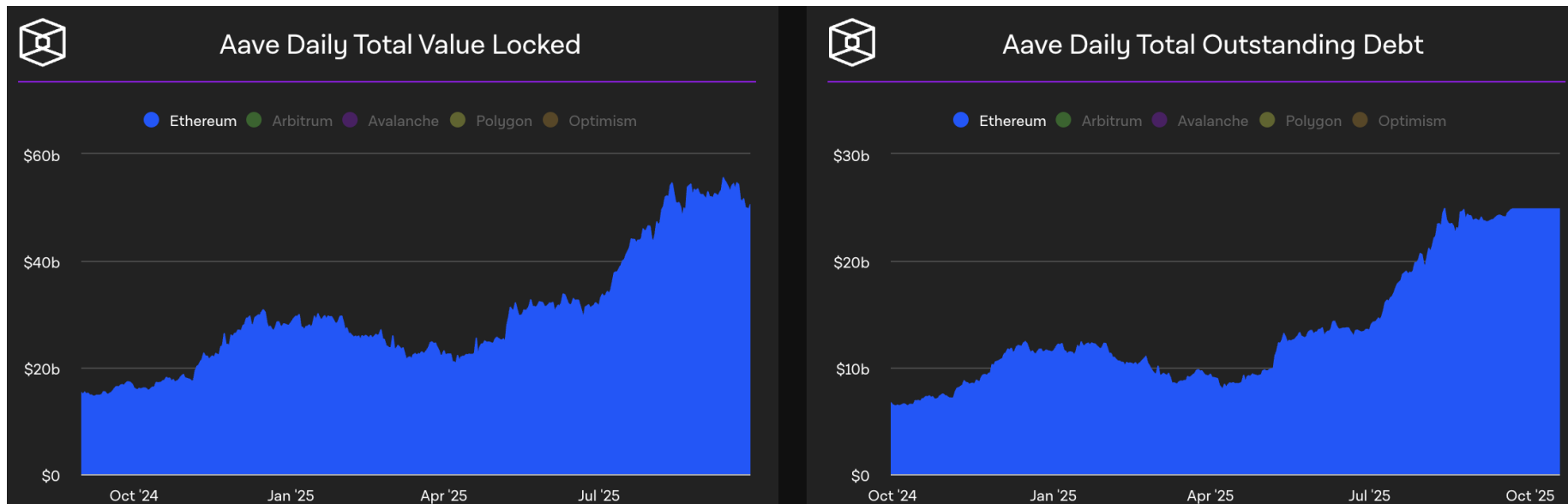
⇒ liquidation ...



Borrowers must constantly monitor the health of their position, and quickly repay loans if it drops (several services automate this)

Summary & stats

- Liquidity providers can earn interest on their assets
- DeFi lending usage (on Ethereum):

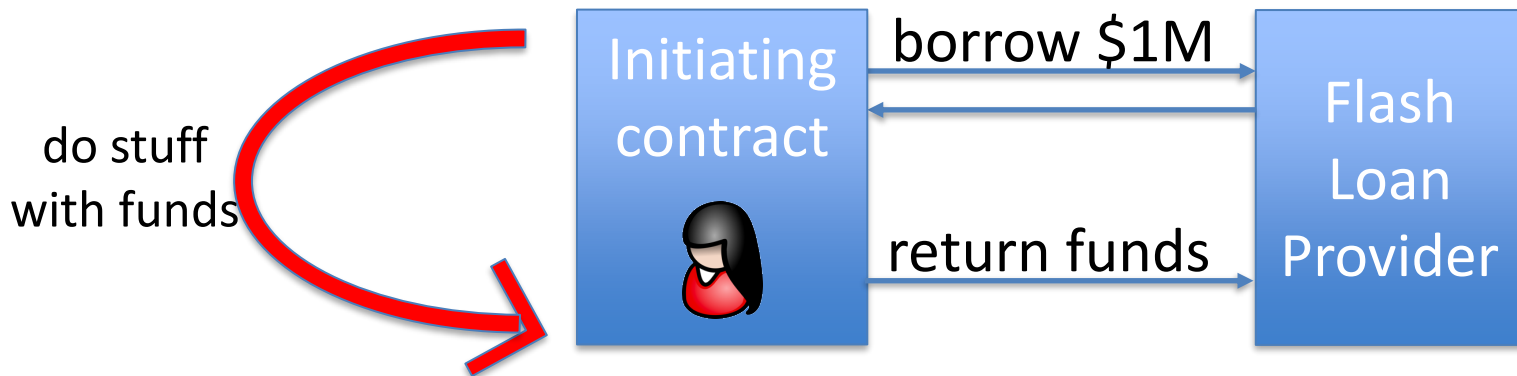


Flash loans

What is a flash loan?

A flash loan is taken and repaid in a single transaction

⇒ zero risk for lender ⇒ borrower needs no collateral



(Tx is valid only if funds are returned in same Tx)

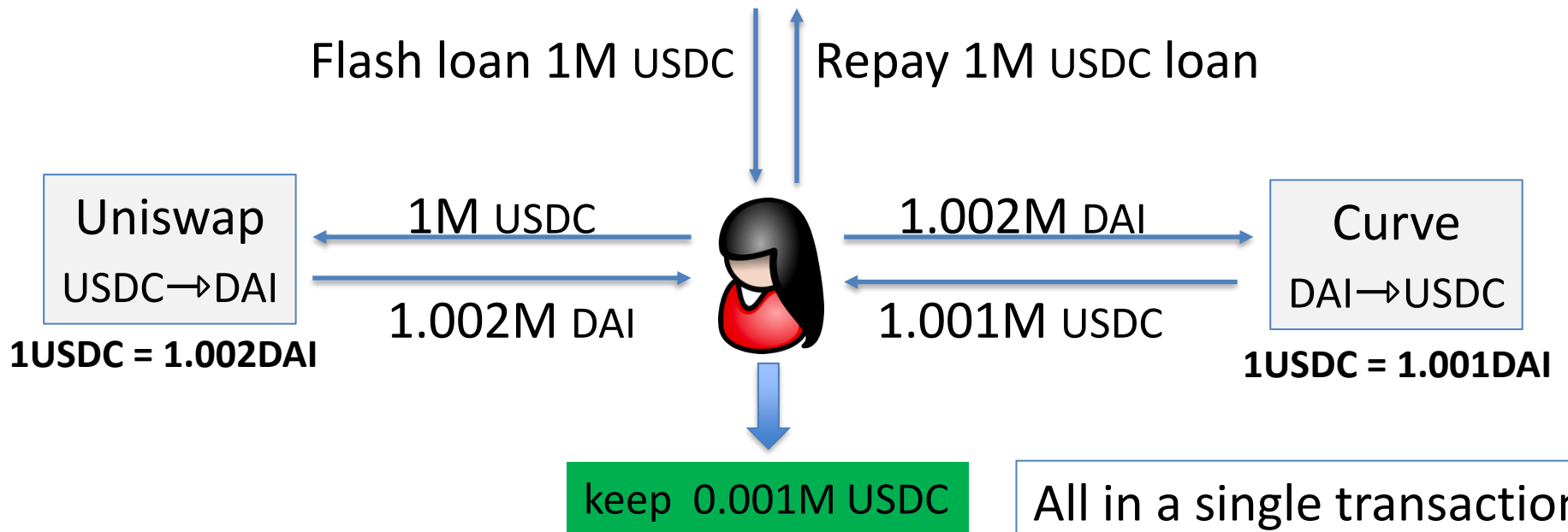
Use cases

- Risk free arbitrage
- Collateral swap
- DeFi attacks: price oracle manipulation
-

Risk free arbitrage

Alice finds a USDC/DAI price difference in two pools

Aave (flash loan provider)



Collateral swap

start:

Alice @Compound



end goal:

Alice @Compound

-1000 DAI
+1 cETH

Take 1000 DAI flash loan
Repay 1000 DAI debt
Redeem 1 cETH
Swap 1 cETH for 3000 cUSDC
Deposit 3000 cUSDC as collateral
Borrow 1000 DAI
Repay 1000 DAI flash loan

-1000 DAI
+3000 cUSDC

borrowed DAI using
ETH as collateral

(a single Ethereum transaction)

borrowed DAI using
USDC as collateral

Aave v1 implementation

```
function flashLoan(address _receiver, uint256 _amount) {  
    ...  
    // transfer funds to the receiver  
    core.transferToUser(_reserve, userPayable, _amount);  
  
    // execute action of the receiver  
    receiver.executeOperation(_reserve, _amount, amountFee, _params);  
    ...  
    // abort if loan is not repaid  
    require( availableLiquidityAfter == availableLiquidityBefore.add(amountFee),  
            "balance inconsistent");  
}
```

END OF LECTURE

Next lecture: Decentralized Exchanges (DeX)